

Galaxy-X-os EM_DASH One-Page Technical Report

SCALE x ODYSSEY | Deep-learning classification of raw astronomical images into 5 celestial categories.

GitHub: <https://github.com/Srujan0798/Galaxy-X-os>

Dataset Sources

- Spiral Galaxy + Elliptical Galaxy EM_DASH real survey imagery from Galaxy10 DECaLS (via astroNN; real SDSS/DECaLS images), mapped from Galaxy10's 10 morphology classes into our two galaxy-morphology classes.
- Nebula + Star Cluster + Planetary Object EM_DASH real imagery from the NASA Image Library (images.nasa.gov / <https://images-api.nasa.gov>, no API key): Hubble / Spitzer / JPL mission images retrieved by keyword and passed through per-class purity filters (e.g. a "planetary nebula" is counted as a nebula and never as a planet; a "galaxy cluster" is never a star cluster). Implemented in `src/download_archives.py`. Kaggle token-authenticated sets (`fedesoriano/deep-space-images`, `brsdincer/planetary-solar-system-objects`) and a clearly-labelled procedural fallback remain as fallbacks only, used for any class the NASA source did not cover at run time.
- 5 classes: Spiral Galaxy, Elliptical Galaxy, Nebula, Star Cluster, Planetary Object.
- Split: stratified disjoint train / val / test. Verified by MD5 hash that no image appears in more than one split (zero train/test leakage).
- Source of truth: the authoritative per-class real-vs-fallback record is `data/processed/DATA_MANIFEST.json`, generated by `src/prepare_data.py`. With the NASA Image Library source working (no API key), all five classes can now be built from real telescope imagery: the two galaxy classes from Galaxy10 DECaLS/SDSS survey images, and Nebula / Star Cluster / Planetary from the NASA Image Library; Kaggle and the labelled procedural generator are fallbacks only. `DATA_MANIFEST.json` records, per class, exactly which source was used and whether it was real.

Model Architecture

- Backbone: EfficientNet-B3 (ImageNet pretrained, transfer learning) EM_DASH ~11.6M parameters.
- Head: Replaced classifier RIGHTWARDS_ARROW 5-class output.
- Training strategy: Progressive unfreezing (backbone frozen first 3 epochs), mixed-precision (`torch.amp`), AdamW (lr 3e-4), batch size 32, OneCycleLR scheduler.
- Augmentations: Albumentations astro-specific pipeline (cosmic-ray simulation, vignetting, Poisson noise, flips/rotations, label smoothing 0.1).
- Explainability: Grad-CAM heatmaps over predictions.

Final Test Metrics

Evaluated on the held-out disjoint test set (249 images, ~50 per class EM_DASH none seen in training). Macro-averaged. All five classes built primarily from real telescope imagery (see `DATA_MANIFEST.json` below); the reported run used real Galaxy10 DECaLS images for the two galaxy classes and real NASA Image Library images for the other three.

Metric	Standard	+ TTA (6× aug)
--------	----------	----------------

-----|-----|-----
-----|

Precision (macro)	0.934	EM_DASH
Recall (macro)	0.932	EM_DASH
F1 (macro)	0.932	0.928

Per-class F1 (Standard):

Class	F1
-------	----

-----|-----|

Elliptical Galaxy	0.895
-------------------	-------

Accur

Spiral

Galaxy-X-os -- SCALE x ODYSSEY Technical Report

Nebula	0.949
Star Cluster	0.947
Planetary Object	0.980

The two real galaxy-morphology classes EM_DASH Spiral vs Elliptical EM_DASH show the realistic ~0.88EN_DASH0.90 F1 that captures genuine morphological ambiguity, which is the residual error. Nebula / Star Cluster / Planetary, now also trained on real NASA imagery (no longer the earlier clean procedural/Kaggle mix), show realistic 0.95EN_DASH0.98 F1 EM_DASH the small drop from the earlier perfect 1.000 reflects the genuine visual variety of real telescope images. TTA is within noise of standard here (the real test set is harder and more varied than the earlier procedural one).

See data/processed/DATA_MANIFEST.json for the honest per-class source breakdown. The committed manifest reflects the real run (Galaxy10 DECaLS for the two galaxies, NASA Image Library for the other three); the small committed preview under data/processed/ is not the full dataset.

> Reproducibility note: the 93.17% / 92.77% numbers were produced on a CUDA GPU (Google Colab, T4) via notebooks/Galaxy_X_Colab.ipynb over a 50-epoch schedule on imagery built primarily from real telescope sources EM_DASH Galaxy10 DECaLS for the two galaxy classes and the NASA Image Library (no API key) for Nebula / Star Cluster / Planetary. The committed data/processed/DATA_MANIFEST.json is the manifest from this real run. To reproduce: run the Colab notebook, or python src/prepare_data.py (downloads real data, no key needed) then python src/train.py on a GPU. See docs/REPRODUCIBILITY.md.

Confusion Matrix

See results/confusion_matrix.png. Additional plots: results/per_class_metrics.png, results/confidence_distribution.png.

Inference Time

- Per-image inference is well under the 5 s requirement on consumer hardware (Apple MPS / CUDA / CPU). Exact latency depends on batch size, device, and whether the model is already loaded; run python src/inference.py <image> --batch-size 1 to measure on your hardware.
- Streamlit demo runs in real time on consumer hardware.

Setup Instructions

```
bash
pip install -r requirements.txt
python src/prepare_data.py      # download + build dataset (writes DATA_MANIFEST.json)
python src/train.py            # train (full EfficientNet-B3 fine-tune)
python src/evaluate.py         # evaluate (standard + TTA)
python src/gradcam.py          # Grad-CAM report
streamlit run app/app.py       # web demo
```

One-click alternative: open notebooks/Galaxy_X_Colab.ipynb in Google Colab to run the full pipeline end-to-end on a free GPU.

Training Script

- Full pipeline: src/train.py EM_DASH progressive unfreezing (backbone frozen epochs 1-3, full fine-tune from epoch 4) + OneCycleLR, AdamW (lr 3e-4), batch 32, mixed precision (AMP), class weights, label smoothing.
- Config: configs/config.yaml. Checkpoint: checkpoints/best_model.pth.

> Training note: the reported model was trained to 93.17% / 92.77% on a CUDA GPU (Google Colab) over a 50-epoch schedule via notebooks/Galaxy_X_Colab.ipynb on imagery built primarily from real telescope sources (Galaxy10 DECaLS + NASA Image Library). Inference runs comfortably on consumer hardware.